Precalculus Review for Calculus

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Functions and composition**

Let $f(x) = \sqrt{x-1}$ and $g(x) = x^2 + 3$. Find:

a $(f \circ g)(x)$ $f(g(x)) = \sqrt{x^2+2}$.

b $(g \circ f)(x)$ $g(f(x)) = (x-1)+3 = x+2$, for $x \geq 1$.

c the domain of $f \circ g$ $x^2 + 2 \geq 1$ always holds, so the domain is all real numbers.

(a) $f(g(x)) = \sqrt{x^2+2}$.; (b) $g(f(x)) = (x-1)+3 = x+2$, for $x \geq 1$.; (c) $x^2 + 2 \geq 1$ always holds, so the domain is all real numbers.

2 **[Foundational]** • **Target: Functions and composition**

Simplify the difference quotient $\frac{f(x+h)-f(x)}{h}$ for $f(x) = x^2 - 3x$.

$\frac{(x+h)^2 - 3(x+h) - (x^2 - 3x)}{h} = \frac{2xh + h^2 - 3h}{h} = 2x + h - 3.$

3 **[Foundational]** • **Target: Lines, exponentials, and trig — the toolkit calculus builds on**

Find the slope and equation of the line through $(-1, 4)$ and $(3, -2)$.

Slope $= \frac{-2-4}{3-(-1)} = -\frac{3}{2}$; equation $y - 4 = -\frac{3}{2}(x+1)$, i.e. $y = -\frac{3}{2}x + \frac{5}{2}$.

4 **[Foundational]** • **Target: Lines, exponentials, and trig — the toolkit calculus builds on**

Solve $2e^{3x} = 14$ exactly and to three decimal places.

$e^{3x} = 7$, so $x = \frac{\ln 7}{3} \approx 0.649$.

5 **[Foundational]** • **Target: Lines, exponentials, and trig — the toolkit calculus builds on**

State the exact value of each expression, needed frequently when differentiating trig functions:

a $\sin\frac{\pi}{4} = \frac{\sqrt{2}}{2}$ **b** $\cos\frac{\pi}{3} = \frac{1}{2}$ **c** $\tan\frac{\pi}{6} = \frac{\sqrt{3}}{3}$

(a) $= \frac{\sqrt{2}}{2}$; (b) $= \frac{1}{2}$; (c) $= \frac{\sqrt{3}}{3}$

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6 [Foundational] • Target: Factoring and rational expressions

Factor each expression fully:

a $x^2 - 5x + 6$ **b** $4x^2 - 9$ **c** $x^3 - 8$

(a) $= (x-2)(x-3)$; (b) $= (2x-3)(2x+3)$; (c) $= (x-2)(x^2+2x+4)$

7 [Foundational] • Target: Factoring and rational expressions

Simplify $\frac{x^2-4}{x^2-x-6}$, stating the value(s) of x excluded from the domain.

$\frac{(x-2)(x+2)}{(x-3)(x+2)} = \frac{x-2}{x-3}$, excluding $x = -2$ (removable) and $x = 3$ (undefined).

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Inverse functions and logarithms

Find the inverse of $f(x) = 3x - 5$, and verify by computing $f(f^{-1}(x))$.

Swap and solve: $x = 3y - 5 \Rightarrow y = \frac{x+5}{3}$, so $f^{-1}(x) = \frac{x+5}{3}$. Check: $f(f^{-1}(x)) = 3 \cdot \frac{x+5}{3} - 5 = x + 5 - 5 = x$. ✓

9 [Intermediate] • Target: Inverse functions and logarithms

Solve $\ln(x-1) = 2$ exactly and to three decimal places.

$x - 1 = e^2$, so $x = e^2 + 1 \approx 8.389$.

10 [Intermediate] • Target: Piecewise and absolute value review

Sketch-reasoning: rewrite $f(x) = |2x - 6|$ as a piecewise function without absolute value bars.

$f(x) = 2x - 6$ for $x \geq 3$, and $f(x) = 6 - 2x$ for $x < 3$ (since $2x - 6 \geq 0$ exactly when $x \geq 3$).

11 [Intermediate] • Target: Piecewise and absolute value review

Solve the inequality $|3x + 1| \leq 7$, writing your answer in interval notation.

$-7 \leq 3x + 1 \leq 7$: $-8 \leq 3x \leq 6$: $-\frac{8}{3} \leq x \leq 2$. Interval: $\left[-\frac{8}{3}, 2\right]$.

12 [Intermediate] • Target: Systems of equations

Solve the system $2x + y = 7$ and $x - y = 2$ by elimination.

Adding the equations: $3x = 9$, so $x = 3$; then $y = 7 - 2(3) = 1$. Solution: $(3, 1)$.

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13 [Intermediate] • Target: Systems of equations

Find the point(s) of intersection of $y = x^2$ and $y = 2x + 3$.

$x^2 = 2x + 3$: $x^2 - 2x - 3 = 0$, $(x - 3)(x + 1) = 0$: $x = 3, -1$. Points: $(3, 9)$ and $(-1, 1)$.

14 [Intermediate] • Target: Graphing functions calculus will build on

The graph of $f(x) = x^3 - 3x$ is shown. Use it to estimate the x-intercepts and the local max/min locations — features calculus will let you find exactly.

From the graph, x-intercepts near $x = 0, \pm 1.7$; local max near $x = -1$, local min near $x = 1$ — these visual estimates match what derivative techniques will later confirm exactly.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Average rate of change (a preview of the derivative) • Cross-Lesson: 5.01 The Fundamental Theorem of Calculus

Find the average rate of change of $f(x) = x^2$ over $[1, 4]$, and interpret this as the slope of a secant line.

$\frac{f(4) - f(1)}{4 - 1} = \frac{16 - 1}{3} = 5$ — this is the slope of the secant line connecting $(1, 1)$ and $(4, 16)$.

16 [Challenging] • Target: Average rate of change (a preview of the derivative) • Cross-Lesson: 6.01 Related Rates

For $f(x) = x^2$, find the average rate of change over $[1, 1 + h]$ in terms of h, and describe what happens as $h \to 0$ (a preview of the derivative).

$\frac{f(1 + h) - f(1)}{h} = \frac{(1 + h)^2 - 1}{h} = \frac{2h + h^2}{h} = 2 + h$. As $h \to 0$, this approaches 2 — the instantaneous rate of change (derivative) of f at $x = 1$.

17 [Challenging] • Target: Radicals and rational exponents • Cross-Lesson: 6.02 Optimization and Curve Sketching

Simplify each expression, writing your answer with positive exponents:

a $x^{1/2} \cdot x^{1/3} = x^{5/6}$ **b** $(x^{2/3})^3 = x^2$ **c** $\frac{x^{5/2}}{x^{1/2}} = x^2$

(a) $= x^{5/6}$; (b) $= x^2$; (c) $= x^2$

18 [Challenging] • Target: Radicals and rational exponents • Cross-Lesson: 7.01 Integration by Substitution

Rewrite $\sqrt[3]{x^2}$ using rational exponent notation, and evaluate it at $x = 8$.

$\sqrt[3]{x^2} = x^{2/3}$. At $x = 8$: $8^{2/3} = (8^{1/3})^2 = 2^2 = 4$.

19 [Challenging] • Target: Interval notation and set operations review • Cross-Lesson: 7.02 Integration by Parts

Write the solution set of $x^2 - 4 > 0$ in interval notation.

$x^2 > 4 \Rightarrow x < -2$ or $x > 2$. Interval: $(-\infty, -2) \cup (2, \infty)$.

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20 [Challenging] • Target: Systems of equations • Cross-Lesson: 8.01 Area Between Curves

Let $F(x) = \int_2^x (t^2 - 1)dt$. According to the Fundamental Theorem of Calculus, what is the derivative of $F(x)$ with respect to x?

The Fundamental Theorem of Calculus, Part 1, states that if $F(x) = \int_a^x f(t)dt$, then $F'(x) = f(x)$. In this problem, $f(t) = t^2 - 1$ and the lower limit of integration is $a = 2$. Therefore, $F'(x) = x^2 - 1$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Functions, Domain, and Composition

Given $f(x) = \sqrt{x - 3}$ and $g(x) = x^2 - 1$, find the domain of $(f \circ g)(x)$.

a. $(-\infty, -2] \cup [2, \infty)$ ✓

b. $[2, \infty)$

c. $[3, \infty)$

d. $[-2, 2]$

$(f \circ g)(x) = \sqrt{g(x) - 3} = \sqrt{x^2 - 4}$. For real output, $x^2 - 4 \geq 0 \Rightarrow |x| \geq 2$, so $x \in (-\infty, -2] \cup [2, \infty)$.

22 [Foundational] • Target: Functions, Domain, and Composition

If $f(x) = \frac{2x+1}{x-3}$, what is $f^{-1}(x)$?

a. $f^{-1}(x) = \frac{3x+1}{x-2}$ ✓

b. $f^{-1}(x) = \frac{x-3}{2x+1}$

c. $f^{-1}(x) = \frac{3x-1}{x+2}$

d. $f^{-1}(x) = \frac{2x-1}{x+3}$

$y = \frac{2x+1}{x-3} \Rightarrow yx - 3y = 2x + 1 \Rightarrow x(y-2) = 3y + 1 \Rightarrow x = \frac{3y+1}{y-2}$. Thus $f^{-1}(x) = \frac{3x+1}{x-2}$.

23 [Foundational] • Target: Functions, Domain, and Composition

Simplify the difference quotient $\frac{f(x+h) - f(x)}{h}$ for $f(x) = 3x^2 - 5x$.

a. $6x + 3h - 5$ ✓

b. $6x - 5$

c. $3x + 3h - 5$

d. $6x + h - 5$

$\frac{3(x+h)^2 - 5(x+h) - (3x^2 - 5x)}{h} = \frac{6xh + 3h^2 - 5h}{h} = 6x + 3h - 5$.

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24 [Foundational] • Target: Exponentials, Logarithms, and Trig Review

Solve the equation $4^{x+1} = 8^{2x-1}$ for x.

a. $5/4$ ✓

b. 1

c. $3/2$

d. $2/3$

Write in base 2: $2^{2(x+1)} = 2^{3(2x-1)} \Rightarrow 2x + 2 = 6x - 3 \Rightarrow 4x = 5 \Rightarrow x = 5/4$.

25 [Foundational] • Target: Exponentials, Logarithms, and Trig Review

Solve $\ln(2x - 1) + \ln(x) = \ln(3)$.

a. $x = 3/2$ ✓

b. $x = 1$

c. $x = 2$

d. $x = 3$

$\ln(2x^2 - x) = \ln(3) \Rightarrow 2x^2 - x - 3 = 0 \Rightarrow (2x - 3)(x + 1) = 0$. Since $x > 1/2$, the only valid solution is $x = 3/2$.

26 [Foundational] • Target: Exponentials, Logarithms, and Trig Review

Find the exact value of $\sin\left(\frac{7\pi}{6}\right) + \cos\left(\frac{4\pi}{3}\right)$.

a. -1 ✓

b. $-1/2$

c. 0

d. $-\sqrt{3}$

$\sin(7\pi/6) = -1/2$ and $\cos(4\pi/3) = -1/2$. Sum $= -1/2 + (-1/2) = -1$.

27 [Foundational] • Target: Rational Functions, Holes, and Asymptotes

Identify the vertical and horizontal asymptotes of $f(x) = \frac{3x^2 - 12}{x^2 - 5x + 6}$.

a. **Vertical:** $x = 3$**; Horizontal:** $y = 3$ ✓

b. Vertical: $x = 2, x = 3$; Horizontal: $y = 3$

c. Vertical: $x = 3$; Horizontal: $y = 0$

d. Vertical: $x = 2$; Horizontal: $y = 1$

Factor: $\frac{3(x-2)(x+2)}{(x-2)(x-3)} = \frac{3(x+2)}{x-3}$ for $x \neq 2$. Removable hole at $x = 2$, vertical asymptote at $x = 3$. As $x \to \infty$, $y \to 3$.

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28 [Foundational] • Target: Rational Functions, Holes, and Asymptotes

Determine the slant asymptote of $g(x) = \frac{2x^2 + 5x - 1}{x + 1}$.

a. $y = 2x + 3$ ✓

b. $y = 2x + 5$

c. $y = 2x - 1$

d. $y = x + 3$

Polynomial long division: $\frac{2x^2 + 5x - 1}{x + 1} = 2x + 3 - \frac{4}{x + 1}$. Slant asymptote is $y = 2x + 3$.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Rational Functions, Holes, and Asymptotes

Find the domain of $h(x) = \frac{\ln(x + 2)}{\sqrt{9 - x^2}}$.

a. $(-2, 3)$ ✓

b. $[-2, 3]$

c. $(-3, 3)$

d. $(0, 3)$

$x + 2 > 0 \Rightarrow x > -2$. $9 - x^2 > 0 \Rightarrow -3 < x < 3$. Intersection gives $x \in (-2, 3)$.

30 [Intermediate] • Target: Trigonometric Identities and Equations

Simplify $\frac{\sin(2\theta)}{1 + \cos(2\theta)}$ into a single trigonometric function.

a. $\tan\theta$ ✓

b. $\cot\theta$

c. $\sec\theta$

d. $\sin\theta$

$\frac{2\sin\theta\cos\theta}{1 + (2\cos^2\theta - 1)} = \frac{2\sin\theta\cos\theta}{2\cos^2\theta} = \frac{\sin\theta}{\cos\theta} = \tan\theta$.

31 [Intermediate] • Target: Trigonometric Identities and Equations

Solve $2\cos^2 x + \cos x - 1 = 0$ on the interval $[0, 2\pi)$.

a. $\pi/3, \pi, 5\pi/3$ ✓

b. $\pi/6, \pi, 11\pi/6$

c. $\pi/3, 2\pi/3$

d. $\pi/2, 3\pi/2$

$(2\cos x - 1)(\cos x + 1) = 0 \Rightarrow \cos x = 1/2$ or $\cos x = -1$. On $[0, 2\pi)$, $x = \pi/3, 5\pi/3$, or π.

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32 [Intermediate] • Target: Trigonometric Identities and Equations

If $\sin\alpha = 3/5$ with $\alpha \in (\pi/2, \pi)$, find $\cos(2\alpha)$.

a. $7/25$ ✓

b. $-7/25$

c. $24/25$

d. $-24/25$

$\cos(2\alpha) = 1 - 2\sin^2\alpha = 1 - 2(3/5)^2 = 1 - 2(9/25) = 1 - 18/25 = 7/25$.

33 [Intermediate] • Target: Rational Functions, Holes, and Asymptotes

The graph of the quadratic function $f(x) = x^2 - 6x + k$ is shown. The vertex of the parabola is at the point $(3, -5)$. Find the value of the parameter k.

a. -5

b. 4 ✓

c. 9

d. 0

The vertex of a parabola in the form $f(x) = ax^2 + bx + c$ occurs at $x = -\frac{b}{2a}$. For the given function $f(x) = x^2 - 6x + k$, we have $a = 1$ and $b = -6$. Thus, the x-coordinate of the vertex is $x = -\frac{-6}{2(1)} = 3$. This matches the given vertex $(3, -5)$. Since the vertex $(3, -5)$ lies on the graph of the function, it must satisfy the function's equation. Substitute $x = 3$ and $f(x) = -5$ into the equation: $-5 = (3)^2 - 6(3) + k$ $-5 = 9 - 18 + k$ $-5 = -9 + k$ To solve for k, add 9 to both sides.

34 [Intermediate] • Target: Trigonometric Identities and Equations

Consider the graph of the function $f(x)$ shown below. Which of the following points lies on the graph of $f(x)$?

a. $(-2, 3)$

b. $(0, -1)$

c. $(1, 0)$ ✓

d. $(3, 4)$

The graph of $f(x)$ passes through the point $(1, 0)$. To verify this, we can observe the graph and identify the coordinates where the curve intersects the x-axis at $x = 1$. Let's check the other options. The point $(-2, 3)$ is not on the curve as the function value at $x = -2$ is approximately 5. The point $(0, -1)$ is not on the curve as the function value at $x = 0$ is approximately 3. The point $(3, 4)$ is not on the curve as the function value at $x = 3$ is approximately 0. Therefore, only $(1, 0)$ lies on the.

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35 [Intermediate] • Target: Functions and composition

Distance between skew lines $L_1 : \mathbf{r} = \mathbf{a} + s\mathbf{d}_1$ and $L_2 : \mathbf{r} = \mathbf{b} + t\mathbf{d}_2$:

a. $|\mathbf{b} - \mathbf{a}|$

b. $|(\mathbf{b} - \mathbf{a}) \cdot (\mathbf{d}_1 \times \mathbf{d}_2)|/|\mathbf{d}_1 \times \mathbf{d}_2|$ ✓

c. $|\mathbf{d}_1 - \mathbf{d}_2|$

d. 0

Applying Vectors and Vector-Valued Functions principles, Skew line distance formula. Therefore, the verified correct option is $|(\mathbf{b} - \mathbf{a}) \cdot (\mathbf{d}_1 \times \mathbf{d}_2)|/|\mathbf{d}_1 \times \mathbf{d}_2|$.

36 [Intermediate] • Target: Lines, exponentials, and trig — the toolkit calculus builds on

What is the maximum rate of change of $f(x, y) = x\cos(y) + y$ at $(-2,\ -2)$?

a. $\sqrt{-4\sin(2) + 2 + 3\sin^2(2)}$ ✓

b. $-4\sin(2) + 2 + 3\sin^2(2)$

c. $-1 - \cos(2) + 2\sin(2)$

d. $-1 + 2\sin(2)$

The maximum rate is $||ablaf||$. Here $ablaf(-2,\ -2) = \langle\cos(2), 1 - 2\sin(2)\rangle$, so it is $\sqrt{-4\sin(2) + 2 + 3\sin^2(2)}$.

37 [Intermediate] • Target: Factoring and rational expressions

What is the maximum rate of change of $f(x, y) = x\cos(y) - 3y$ at $(2,\ 3)$?

a. $3\sin^2(3) + 12\sin(3) + 10$

b. $2\sin(3) - \cos(3) + 3$

c. $\sqrt{3\sin^2(3) + 12\sin(3) + 10}$ ✓

d. $2\sin(3) + 3$

The maximum rate is $||ablaf||$. Here $ablaf(2,\ 3) = \langle\cos(3),\ -3 - 2\sin(3)\rangle$, so it is $\sqrt{3\sin^2(3) + 12\sin(3) + 10}$.

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Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Inverse functions and logarithms • Cross-Lesson: 7.01 Integration by Substitution

What is the maximum rate of change of $f(x, y) = x\cos(y) - 2y$ at $(3, -2)$?

a. $\sqrt{-12\sin(2) - 4\cos(4) + 9}$ ✓

b. $-12\sin(2) - 4\cos(4) + 9$

c. $-2 - \cos(2) + 3\sin(2)$

d. $-2 + 3\sin(2)$

The maximum rate is $||ablaf||$. Here $ablaf(3, -2) = \langle\cos(2), -2 + 3\sin(2)\rangle$, so it is $\sqrt{-12\sin(2) - 4\cos(4) + 9}$.

39 [Challenging] • Target: Piecewise and absolute value review • Cross-Lesson: 7.02 Integration by Parts

Consider the function $f(x) = 2x^2 - 3x + 1$. What is the value of $f(3)$?

a. 10

b. **13** ✓

c. 16

d. 19

To find the value of $f(3)$, we substitute $x = 3$ into the function $f(x) = 2x^2 - 3x + 1$. $f(3) = 2(3)^2 - 3(3) + 1$ $f(3) = 2(9) - 9 + 1$ $f(3) = 18 - 9 + 1$ $f(3) = 9 + 1$ $f(3) = 10$. Therefore, $f(3) = 10$.

40 [Challenging] • Target: Systems of equations • Cross-Lesson: 8.01 Area Between Curves

A spherical balloon is inflated at 4 cm³/s. Rate of radius change when $r = 2$:

a. $\frac{1}{\pi}$

b. $\frac{1}{2\pi}$

c. $\frac{1}{4\pi}$ ✓

d. $\frac{1}{8\pi}$

From $V = \frac{4}{3}\pi r^3$, we have $\frac{dV}{dt} = 4\pi r^2 \frac{dr}{dt}$. At $r = 2$ and $\frac{dV}{dt} = 4$, $\frac{dr}{dt} = \frac{4}{4\pi(4)} = \frac{1}{4\pi}$ cm/s.

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41 [Challenging] • Target: Graphing functions calculus will build on • Cross-Lesson: 8.02 Volumes of Revolution

What is the maximum rate of change of $f(x, y) = x^2e^y + y$ at $(2, \ -1)$?

a. $\frac{16 + (e + 4)^2}{e^2}$

b. $\frac{2 \cdot 2^{\frac{13}{476}} \cdot 3^{\frac{621}{952}} \cdot 5^{\frac{577}{952}} \cdot 7^{\frac{447}{952}}}{7}$

c. $\frac{\sqrt{16 + (e + 4)^2}}{e}$ ✓

d. $1 + \frac{4}{e}$

The maximum rate is $||ablaf||$. Here $ablaf(2, \ -1) = \langle \frac{4}{e}, 1 + \frac{4}{e} \rangle$, so it is $\frac{\sqrt{16 + (e + 4)^2}}{e}$.

42 [Challenging] • Target: Average rate of change (a preview of the derivative) • Cross-Lesson: 8.03 Arc Length and Improper Integrals

Symmetric equations of line through (x_0, y_0, z_0) with direction (a, b, c):

a. $ax + by + cz = 0$

b. $x/a = y/b = z/c$

c. $\frac{x - x_0}{a} = \frac{y - y_0}{b} = \frac{z - z_0}{c}$ ✓

d. $x + y + z = 1$

Applying Vectors and Vector-Valued Functions principles, Symmetric form. Therefore, the verified correct option is $\frac{x - x_0}{a} = \frac{y - y_0}{b} = \frac{z - z_0}{c}$.

43 [Challenging] • Target: Radicals and rational exponents • Cross-Lesson: 9.01 Separable Differential Equations

A 1000-L tank holds brine with 10 kg of salt. Brine with $\frac{1}{2}$ kg/L enters at 5 L/min and the well-mixed solution leaves at the same rate. How much salt (kg) is in the tank after 30 min?

a. $500 - \frac{500}{e^{\frac{3}{20}}}$

b. 85

c. $\frac{10}{e^{\frac{3}{20}}} + 500$

d. $500 - \frac{490}{e^{\frac{3}{20}}}$ ✓

$A' = \frac{5}{2} - \frac{5}{1000}A$, a linear equation with equilibrium 500. So $A = 500 - 490e^{-\frac{t}{200}}$ and $A(30) = 500 - \frac{490}{e^{\frac{3}{20}}}$.

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44 **[Challenging]** • **Target: Interval notation and set operations review** • **Cross-Lesson: 10.01 Calculus with Parametric and Polar Curves**

A 1000-L tank holds brine with 10 kg of salt. Brine with $\frac{1}{10}$ kg/L enters at 10 L/min and the well-mixed solution leaves at the same rate. How much salt (kg) is in the tank after 30 min?

a. $100 - \frac{90}{e^{\frac{3}{10}}}$ ✓

b. $100 - \frac{100}{e^{\frac{3}{10}}}$

c. 40

d. $\frac{10}{e^{\frac{3}{10}}} + 100$

$A' = 1 - \frac{10}{1000}A$, a linear equation with equilibrium 100. So $A = 100 - 90e^{-\frac{t}{100}}$ and $A(30) = 100 - \frac{90}{e^{\frac{3}{10}}}$.

45 **[Challenging]** • **Target: Functions, Domain, and Composition** • **Cross-Lesson: 11.01 Sequences and Series**

A 1000-L tank holds brine with 10 kg of salt. Brine with $\frac{1}{5}$ kg/L enters at 25 L/min and the well-mixed solution leaves at the same rate. How much salt (kg) is in the tank after 30 min?

a. $200 - \frac{200}{e^{\frac{3}{4}}}$

b. 160

c. $\frac{10}{e^{\frac{3}{4}}} + 200$

d. $200 - \frac{190}{e^{\frac{3}{4}}}$ ✓

$A' = 5 - \frac{25}{1000}A$, a linear equation with equilibrium 200. So $A = 200 - 190e^{-\frac{t}{40}}$ and $A(30) = 200 - \frac{190}{e^{\frac{3}{4}}}$.

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